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Development of School Bus Security System Based on RFID and GSM Technonologies for Klang Valley Area

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***Abstract*— At present, school bus services play a vital role to transport students in every country region from home to schools. Many of parents are relying very much on the school bus services although they are worried about their children safety. Realizing of that problem, this paper proposed a SMS based solution to assist most of parents to identify their kid's movement (in/out) of the school bus. SMS will be sent to parents once their children ride on the bus and reached the destination. The project aims to develop a school bus security system using RFID and GSM technologies. RFID is used to identify the student's identity with the parent's contact number and also head count whereas GSM is used as a platform to notify parent on their kid's movement via SMS. Tests has been conducted on the overall performance of the developed prototype. From the results, it was found that the developed project is capable to provide a real-time messaging system to parents on their children whereabouts; with an additional feature of student attendance checker. In brief, this developed prototype system could give the peace of mind for most of parents who are entrusting their children traveling with the school bus transportation.**

Keywords*— *RFID RC-522, SIM-800L GSM module, Arduino, Real Time

I. INTRODUCTION

Nowadays, most of parents are relying on school bus services to transport their children to schools. There are quite a common anxiety feelings have risen among the parents about their children's safety when traveling by school bus. In facts, many of parents suffered a problem whereby the school bus driver forgetting their kids on board. Indirectly, this subject may lead to the increase of kidnapping crime activity among the school's students. Statistical report by national crime bureau states that in every eight minutes children go missing caused by kidnapping crime activities [1].

A number of researchers have conducted the amount of work related to the RFID based security system. An article [1] presented the school bus tracking system based on RFID technology and GSM module. The RFID tag was embedded inside the student identity card in order to provide the tracking mechanism. The notification messages will be sent to the parents throughout the SIM-300 GSM module technology. The major problem with this framework is that the system used the older version of GPRS functionalities. In other research, the Kindergarten Intelligence Security has utilized the used of microcontroller as an interface to interact with RFID and GSM modem. The embedded RFID tags will be placed inside the children uniform in order to provide the security monitoring [2]. The system is design based on the IOT (Internet of Thing) framework. Hence, all the information regarding the students will be stored in a cloud computing services. Child security system also proposed by [3].

It has been proven that the RFID technology is suitable and effective to be used for the security purposes. RFID is very well known with its application of multilevel car parking control and door control for different access of users [4-5]. RFID has also been proposed for halal industry for checking and recording purposes [6]. As a summary, RFID technology is proven to be able to work for real-time tracking and security monitoring.

Realising the importance of school bus security, this project aims to develop a security system that is possibly implemented at every school busses registered in Klang Valley area. The focus group is all school children from kindergarten to secondary. This security system comprises with the technology of Radio Frequency Identification (RFID), Global System for Mobile communication (GSM), and the Arduino microcontroller. Besides user identification, RFID technology is used to count every entry and exit of the student's movement from the school buses. Meanwhile, GSM technology is used to allow the communication with the receiver (parents). The notification message will be sent to the parents by using the Short Message Service (SMS) technique. Besides, the details of students departed or arrived will be included in the notification messages. The implementation of the designed project reduce could give peace of mind to the parents who are entrusting their children traveling with the school bus, reduce anxiety and possibly would create more effective working environment. Klang Valley has been chosen as the focus area due to sustainability of GSM and high dependency on school bus services from working parents.

II. Design of School Bus Security System

This paper proposed the system model that suitable for implementing the safety mechanism for the school bus transportation where attendance is taken and parents are informed via SMS. A system that count every entry and exit of students based on recognizing the unique identification information of the RFID technology has been performed. Once a student touch his/her RFID tag to the RFID reader, the system will display each of student's names that have been registered on the Liquid Crystal Display (LCD) screen module. The presence of the GSM module technology enables most of parents to be informed while their children is entering or leaving the school bus via the SMS. The operation mechanism between the components has been controlled by a microcontroller, Arduino Uno board. Fig. 1 illustrates how the school bus security system works.

Fig. 1. Proposed School Bus Security System

A. Workflow Diagram

The design procedure for the developed system begins with the installation of RC-522 RFID reader in every school bus entrance or exit doors. Students are provided with the passive tags that have been registered with unique serial identification number. RC-522 RFID reader will match the unique identification number of the respective students and display it on the LCD screen module. Meanwhile, GSM technology provides a response to the given data and send the notification message towards the registered parent's mobile phone number. Two different timeframes have been set (morning and afternoon) to indicate the student is on board from home/school or otherwise. This feature is useful to alert the parents about their children attendance. Fig. 2 shows the workflow diagram of the proposed system.

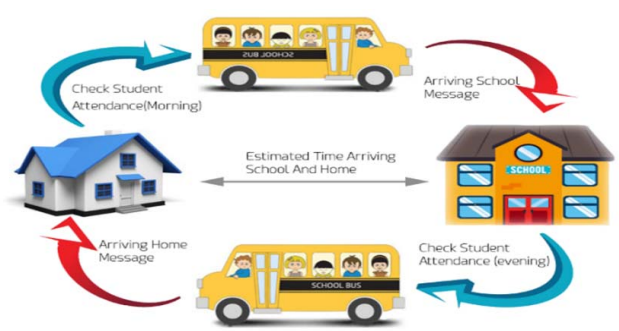


Fig. 2. Workflow Diagram of School Bus Security System

B. Flowchart of School Bus Security System

Fig. 3 illustrates the flowchart of the design system to check the attendance of student during morning and evening session. While Fig. 4 shows the flowchart of the design system sending a real-time notification message when the student has safely arrived at the respective location.

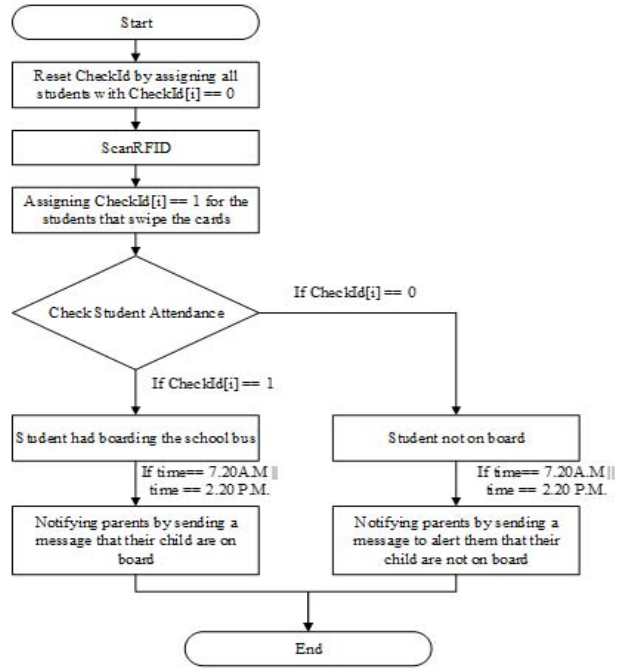


Fig. 3. Flowchart of Attendance Checker

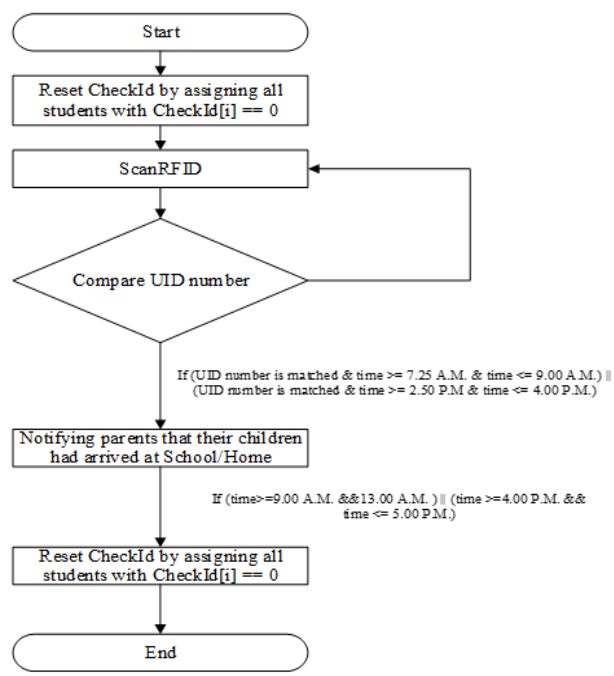
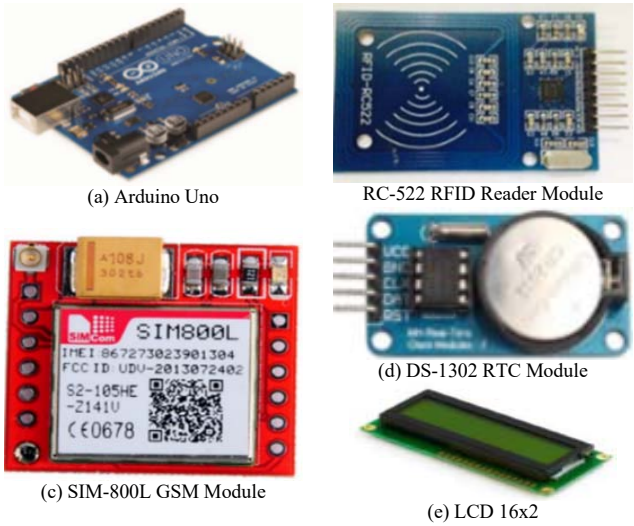


Fig. 4. Flowchart of Real Time Notification

III. IMPLEMENTATION OF THE PROPOSED SYSTEM

In this section, hardware and software implementation of the proposed school bus security system will be elaborated. Each hardware component will be discussed in details, as shown in Fig. 5. Moreover, software implementation using Arduino IDE will be discussed.



(a) Arduino Uno

RC-522 RFID Reader Module

(c) SIM-800L GSM Module

(d) DS-1302 RTC Module

(e) LCD 16x2

Fig. 5. Hardware Components of the Proposed System

A. Arduino Uno

The Arduino Uno, as shown in Fig. 5(a), has been used as the main circuit to control the input and output of the system. It also operates as a CPU to govern the operation between another components. Besides, the circuitry board can be easily to program and reprogram compared to the other microcontrollers likes ATMEL and PIC. This physical board has been built along with 2Kbytes SRAM and 32Kbytes of Flash memory technology [2][9].

B. RC-522 RFID Reader Module

RC-522 RFID module, as shown in Fig. 5(b), is a device that able to transfer or read the data over the short distance. This RFID reader module having a capability to read over 0 to 10mm distance range. Besides, this RC-522 RFID tag is predefined by a unique ID number that can be read by the RFID reader. The communication between the RFID reader and tag takes place by using 13.56MHZ electromagnetic field. Using the same frequency, Halal Kit identifier has been proposed earlier [6]. This RFID RC-522 reader need to be supplied with 3.3V of Arduino Uno board in order to decode the inquiry signal from the tag and sending a respond towards the microcontroller.

C. SIM-800L GSM Telecommunication Module

The SIM-800L GSM module, as shown in Fig. 5(c), is a telecommunication device that used to send and receive a message from the user. By inserting the SIM card in a providing slot, it acts similarly as a mobile phone network. Besides, the presence of GSM modem integrated within the module enables it to operate over the wireless network. This concept has been used to develop digital wireless meter [7]. However, this GSM module needs to initialize its signal coverage before it can operate with the Arduino Uno microcontroller. The suitable operating voltage for this module to send a notification message is about 3.7V to 4.2V.

D. DS-1302 Real Time Clock (RTC) Module

Real Time Clock (DS-1302), as shown in Fig. 5(c), is a module that having a functionality to store time and date in single chip circuit. This single chip circuit integrated inside the DS-1302 RTC module is known as Timekeeping chip. Besides, this chip embedded with a Trickle-charger circuitry [8]. The purpose of having a Trickle-charger circuitry is to ensure the capacitor can maintain the operation of clock while the power supply (VCC) is absent. The frequency of RTC oscillator is about 32.768 KHz and having a pull-up resistor up to 2.2 KΩ. This module only consuming power about less than 1µW.

E. 16x2 Liquid Crystal Display (LCD)

A 16x2 Liquid Crystal Display (LCD), as shown in Fig. 5(e), is used to display the status information of a student either in/out from the school bus.

F. Software Implementation using Arduino IDE

The Arduino IDE (Integrated Development Environment) is a software that aids the programmer in the development of the codes. There are a lot of toolbar and menu features for sketching the codes of the desired system. This software allows the new program that has been sketched to be uploaded into the Arduino board.

IV. RESULTS AND DISCUSSION

A. Overall Performance

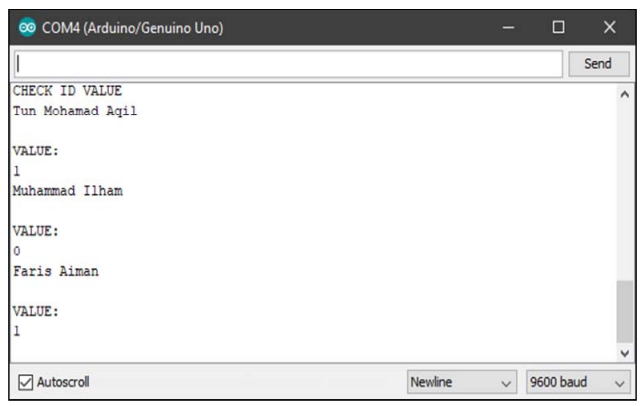


Fig. 6. Serial Monitor Output After Scanning Operations

The serial monitor of the Arduino IDE is used to display an information regarding the check ID of the respective students. Initially, the developed school bus security system had assigned all the students that have been registered with a check ID equivalent to '0'. Students have been asked to scan their RFID tags at the RC-522 RFID reader module. Once the UID number of the students was matched with the registered information system, the value of the check ID will turn into '1'.

The purpose of assigning the check ID value was to indicate the attendance of the students on the school bus. The value of check ID equivalent to '0' means that the students was absent or not boarding the school bus, while the value of check Id equal to '1' means the students had attended or boarding the school bus. Fig. 6 shows the output from the serial monitor after scanning operation.

B. Overall Implementation

The design system implemented with four different student unique identification cards. First, the system need to be supplied with a 9V power adapter. The purpose of having 9V power adapter is to ensure that the system able to initialize the SIM-800L GSM module. The initialization of GSM module need to be performed before the system began the operation. Once the system has been activated, the LCD screen displayed the instruction to scan the RFID tag including with the real time clock. Fig. 7 shows the interface of the develop system.

Fig. 7. The Interface of the Proposed System

Fig. 8. LCD Output of Attendance Checker during Morning Session

There are four possibilities of the output that had been displayed on the LCD screen module as well as in the parents' mobile phones. OUTPUT 1 (Morning Attendance): During the morning session, the school bus driver manages to pick up students at each of their respective house. The programmed read the initial timeframe in order to notify the parents either their kids had attended or absent to the school session. Fig. 8 indicates the output from the LCD module to check the overall attendance of the students during the morning session.

The design system model checked the overall attendance of the students and sent a notification message in the bulk form based on the timeframe given. Fig. 9 shows the notification message to indicate the attendance of the students during the morning session.

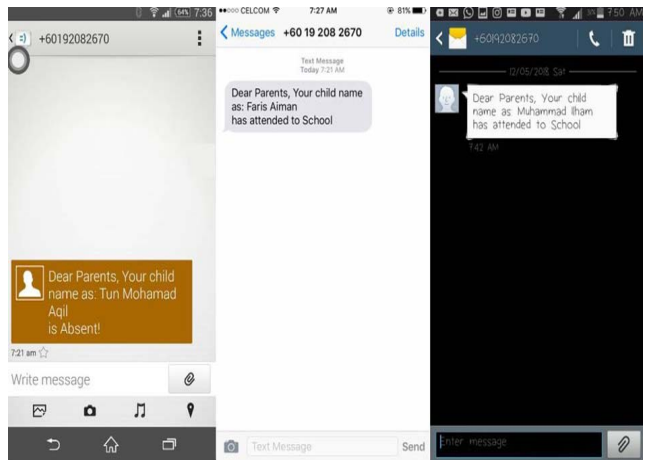


Fig. 9. Real Time Attendance Notification to Parents

As the students leaving the school bus, the LCD screen module printed out the name of each students that have been successfully arrived at the school. Fig. 10 shows the output from LCD screen module indicating the students have successfully arrived at the school compound.

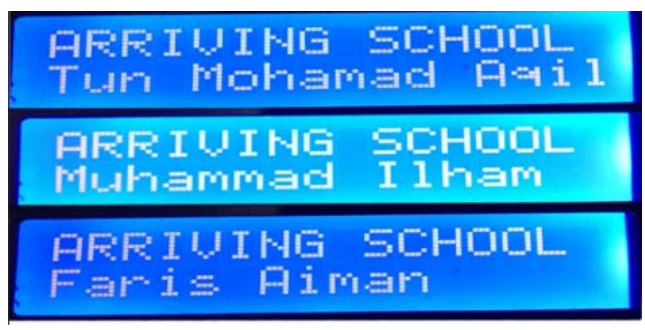


Fig. 10. LCD Output Indicating Students Arrival at School

A notification message directly will be sent to the parents when the students swiped their cards on the particular RFID reader. Fig. 11 shows a real time message indicating the students have arrived at the school compound. Similarly, Fig. 12 shows a real time notifications to parents that the particular students have boarded the school bus, while Fig. 12 shows a

real time message indicating that the particular student has arrived at home.

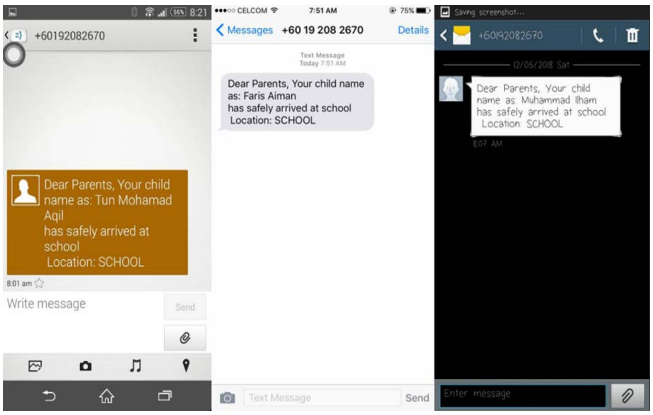


Fig. 11. Real Time Arrival Notifications to Parents

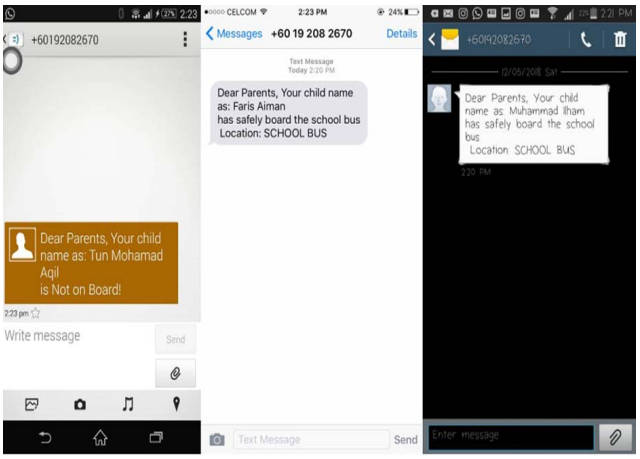


Fig. 12. Real Time Notifications to Parent when Particular Student has Boarded the School Bus in the Afternoon

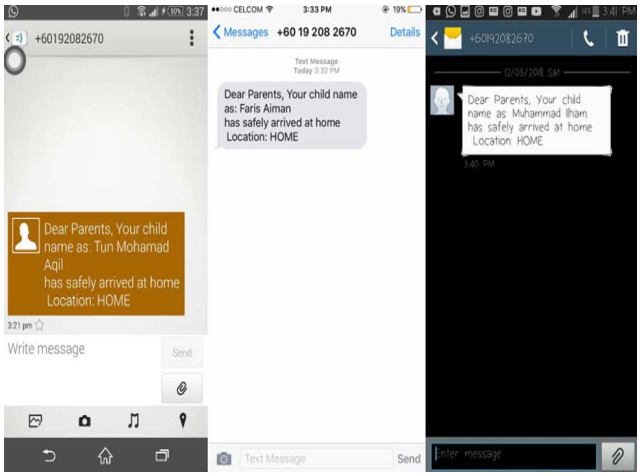


Fig. 13. Real Time Message Indicating the Particular Student has Arrived at Home

V. Conclusions and Future Works

This paper has proposed and implemented a school bus security system based on RFID and GSM technologies. The implementation of the RFID technology could detect the student's movement entering or leaving the school buses. The notification message is sent directly to the parent mobile's phone as soon as their children were successfully departed or arrived to the destination and give piece of mind to the parents. Attendance checker, as an additional feature will give early notification to the parents if the children is not on board (missing). In addition, the proposed project can be further improvised by implementing the Global Positioning System (GPS) as the location tracker. In terms of size and convenient, wristband with a coil-on chip technology could give more advantage than using the student identity card. However, for the long run implementation the effectiveness of the proposed system model can be evaluated based on the prevention measurement of the kidnapping crime activities among the school age children.

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